



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

Follow For
More Updates



COMMON HALF YEARLY EXAMINATION - 2024 - 25

SCIENCE

Time Allowed : 3.00 Hours

[Max. Marks : 75]

12x1= 12

I. Choose the correct answer.

- 1 1 Metric ton is equal to
 - a) 100 Quintals
 - b) 10 Quintals
 - c) $1/10$ Quintals
 - d) $1/100$ Quintals
- 2 Clouds float in atmosphere because of their low _____
 - a) Density
 - b) Pressure
 - c) Velocity
 - d) Mass
- 3 Which of the following converts electrical energy into mechanical energy?
 - a) Motor
 - b) Battery
 - c) Generator
 - d) Switch
- 4 A ray of light passes from one medium to another medium. Refraction takes place when angles of incidence is
 - a) 0°
 - b) 45°
 - c) 90°
 - d) 60°
- 5 _____ is essential to perform separation by solvent extraction method.
 - a) Separation funnel
 - b) Filter paper
 - c) Centrifuge machine
 - d) Sieve
- 6 Modern periodic law states that the physical and chemical properties of elements are the periodic functions of their _____.
 - a) Atomic Numbers
 - b) Atomic Masses
 - c) Similarities
 - d) Anomalies
- 7 Number of valence electrons in Carbon is
 - a) 2
 - b) 4
 - c) 3
 - d) 5
- 8 Which of the following is a complex tissue?
 - a) Parenchyma
 - b) Collenchyma
 - c) Xylem
 - d) Sclerenchyma
- 9 Stomach of human beings mainly digests _____.
 - a) Carbohydrates
 - b) Proteins
 - c) Fat
 - d) Sucrose
- 10 Which of the following is transmitted through air?
 - a) Tuberculosis
 - b) Meningitis
 - c) Typhoid
 - d) Cholera
- 11 Chlorophyll in a leaf is required for _____.
 - a) Photosynthesis
 - b) Tropic Movement
 - c) Transpiration
 - d) Nastic movement
- 12 A presentation consist of many _____.
 - a) Pages
 - b) Slide
 - c) Place holders
 - d) Messages

II. Answer any seven (7) questions only Question no 22 is compulsory.

7x2=14

13. Differentiate Mass and Weight?

14. State Pascal's Law/

15. Match the following

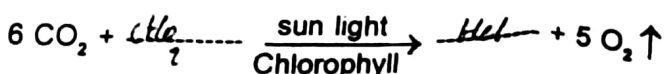
- | | |
|-------------------------|--------------------|
| a) Electric charge | - Ohm <i>c</i> |
| b) Potential difference | - amperes <i>d</i> |
| c) Resistance | - Coulomb <i>b</i> |
| d) Electric current | - Volt <i>a</i> |

16. Define Sublimation.

17. Write a note on different types on Bonds.

18. What are the uses of 'Plaster of Paris'.

19. Fill in the Blanks.



20. **Differentiate:** Incisors and Canines

21. Explain any two methods of Food preservation

22. Calculate the amount of charge that would flow in 2 hours through an element of an electric bulb drawing a current 2.5 A

III. Answer any 7 questions: (Q.No.32 is compulsory)

7x4=28

23 i) Define Least count of any device

ii) Define Velocity

24 i) Name any two appliances which work under the Principle of heating effect of Current

ii) How are the Home Appliances connected in General, in series or Parallel Give Reason

25 i) What is the Speed of Light in Vacuum?

ii) What is meant by Magnification? Write its expression.

26 i) Match the following

a) Dalton - i) Hydrogen atom model

b) Chadwick - ii) Discovery of Nucleus

c) Rutherford - iii) First Atomic theory

d) Neils Bohr - iv) Plum Pudding model

v) Discovery the neutrons

ii) What are the limitations of Mendeleev's Periodic table?

27 i) Find the Oxidation number of the elements in the following compounds.

a) C in CO_2

b) Mn in MnSO_4

ii) Define Aquaregia

28 i) True or False? (If False give the correct statement)

a) Fishes have two chambered heart. b) What is Complex tissue?

ii) Skin of reptilians are smooth and moist.

29 Difference between tropic and Nastic Movement? any four?

30 Explain the Structure of Nephron

31 i) Match the following.

a) Swine flu - Human Papilloma virus

b) Genital warts - Human Immuno deficiency virus

c) AIDS - Mycobacterium

d) Tuberculosis - Influenza virus H_1N_1

ii) What is a Slide?

32 Write the Electronic configuration of 'K' and 'Cl'

IV. Answer all the question.

3x7=21

33 a) (i) Explain different types of Motion. (any five)

(ii) State the Laws of Flotation? (OR)

b) (i) Explain the principle, construction and working of a DC. Motor.

34 a) How is a mixture of common salt, Oil and Water separated? You can use a combination of different methods. (OR)

b) i) List down the differences between ionic and covalent compounds.

ii) Sulphuric acid is called king of Chemicals. Why is it called so?

35 a) i) List out the differences between Mitosis and Meiosis.

ii) Why are Frogs said to be Amphibians.

(OR)

b) How are Vitamins useful to us? Tabulate the Sources, deficiency diseases and Symptoms of Fat Soluble Vitamins



SATGURU STUDY CENTRE

17/8, Poondi Thangammal St, Nagoor Garden, Tondiarpet, Chennai, Tamil Nadu 600081
An MSME-Certified Learning Centre

CH / 9 / Sci / 2